

TIADC Calibration Loop Structure and Error Root-Cause Analysis

1. Scope and Approach

This report analyses the structure and error sources of the TIADC calibration loop. Three classes of problems are distinguished:

- 1. **Algorithmic / fundamental limitations:** dither three-way separation, link dispersion, flat-top assumption;
- 2. **Link / hardware limitations:** DAC-to-ADC analog bandwidth, clock jitter, actuator resolution;
- 3. **Control / implementation issues:** stage failure propagation, state invalidation, gain self-normalisation, closed-loop rejection logic.

The report draws on board measurements, desktop simulation, and production-flow reproductions on a simulated board.

2. Calibration Loop Structure

2.1 Five-Stage Firmware Pipeline

The firmware calibration flow (`adc -cal`) consists of five serial stages:

Stage	Main task	Pass condition	Effect of failure
Prepare (implicit)	Reference upload, actuator neutral initialisation, warm-up	reference/readiness checks pass	Timing is marked failed
1 Timing	Reference alignment, cross-correlation, lag/fractional lag, channel selection	alignment quality and correlation thresholds pass	Timing context invalidated; later stages cannot run
2 Offset	Tone DC fit + closed-loop offset correction	offset converges and verifies	Gain input invalidated; later stages cannot run

Stage	Main task	Pass condition	Effect of failure
3 Gain	Tone amplitude fit + closed-loop gain correction + verification	gain converges, verification passes, output valid	Gain input invalidated
4 Skew	Tone phase-difference closed loop + dither advisory check	skew measurement/correction succeeds	Calibration fails
5 Performance	SNDR/SFDR/THD/ENOB, A/B matching	spectrum/matching valid	Marked unavailable; no rollback of earlier corrections

Key structural properties:

- The five stages are strictly serial; a failing stage blocks later stages;
- An unusable Offset result clears the Gain input;
- Skew requires a usable correction output from the first three stages;
- Performance is the last stage that may fail; it does not roll back earlier corrections.

2.2 Host-Side Closed Loop

One host-side loop iteration is:

```
capture → de-frame → apply current correction → estimate residual
→ block-LMS update → push skew to the clock-delay actuator → log
```

Design characteristics:

- **Correction is applied before residual estimation**, so the recorded trajectory is a true closed-loop learning curve;
- **Skew is written back through the AD9695 fine-delay register**;
- Frame-level rejection logic (alignment margin, event count, finiteness, skew/gain ranges);
- Capture retries to prevent bad frames from entering the LMS;
- Depends on UART/UDP capture; network timing affects loop speed.

2.3 Estimator Structure

Branch	Principle	Current status
Tone primary	800-sample DC+cos+sin least-squares per frame; A/B phase difference	Stable; <1 ps on the board
Dither offset	Polarity-weighted event averaging + flat-top sampling	Weak pass on board; simulation shows the flat-top assumption is not valid
Dither gain	Least-squares projection onto the template (full/flat)	No constant calibration factor
Dither skew	Polarity-weighted event aggregation → A template → B fit → derivative projection	Not reliable on board; usable only in ideal simulation
Joint aggregation	Cross-frame stacking of event windows	Not reliable on board; occasional offline pass

3. Error Categories and Root Causes

3.1 System / Link-Level Causes

(1) DAC-to-ADC link dispersion

- Observation: a 32–48 sample digital pulse arrives as a **74–123 sample-wide** pulse at the ADC (10–90% width);
- Cause: DAC zero-order hold and reconstruction filtering, finite analog-injection bandwidth;
- Effect: the flat top is not preserved, edges are rounded, and event localisation / template projection becomes sensitive.

(2) DAC/ADC sample-rate ratio constraint

- The dither must land on a fixed ADC sample phase, so f_{s_dac} / f_{s_adc} must be an integer;
- A non-integer ratio smears the averaged pulse replica and directly biases the gain estimate.

(3) Tone-fit pollution

- Dither and tone share the same frame; pulse energy enters the tone-fit residual;
- A larger pulse duty cycle increases batch noise (baseline std 2–9 → 23 ps).

(4) Jitter and analog bandwidth at high frequency

- Phase noise scales linearly with frequency;
- ADC input bandwidth and aperture uncertainty become more significant at high frequency;
- The algorithm may remain accurate, but the analog front end can become the limit.

3.2 Timing-Stage Errors

Symptom	Cause
Timing fails	Reference not uploaded, length/format mismatch, correlation below threshold
Lag anomaly	Arbitrary reference start phase, uncertain channel selection, poor fractional-lag condition
Everything downstream fails	Timing failure invalidates the timing context, so Offset/Gain/Skew cannot start
Repeated frames	Port/firmware mismatch, UDP packet loss, capture not aligned with reference

3.3 Offset-Loop Errors

(1) The original estimator relies on a "true flat top"

- It searches the digital template for a region with near-zero derivative and near-peak amplitude;
- That region is defined in the digital domain, not measured at the ADC;
- After dispersion, those samples lie on slopes, so the offset estimate is contaminated by shape and localisation errors.

(2) Window DC / slow-varying bias

- A/B window means correlate with polarity (about +1.8/−1.6 codes);
- Rising/falling masks are asymmetric, producing opposite edge biases of ~100–240 ps.

(3) Event-centre jumps and insufficient event count

- In-frame phase jumps (44→9) and spacing outliers;
- With few events, averaging noise is high and polarity balance degrades.

(4) Controller accuracy

- The offset loop converges, but dither offset deviates 3–4 codes from tone DC;

- This is a weak in-tolerance pass, not evidence that dither independently estimates offset accurately.

3.4 Gain-Loop Errors

(1) The production gain loop is self-normalising

- It applies the current correction before measuring, so a constant channel-gain error is invisible to the loop;
- The correction stays at 1.0 and the mismatch appears only in the nominal system gain.

(2) Dither-gain template-domain to ADC-code-domain mapping

- Board dither gain is 0.24–0.44 against a true value of 1.0;
- flat $\approx$ full (about 0.249 vs 0.235);
- Both measure the same dispersion-rounded pulse, and no physical flat top exists;
- There is no constant ratio across runs, so online absolute calibration is not possible.

(3) Fit quality and event count

- Dither-gain valid rate is about 2/3;
- Few events, template pollution, and poor fit quality lead to WARNING/INVALID.

3.5 Skew-Loop Errors

(1) Tone primary path

- Board convergence: **–0.87 ps, 2/2 pass**;
- Loop structure: characterise actuator $\rightarrow$ step by batch median $\rightarrow$ converge with consecutive passes;
- Main limits are actuator resolution and run-to-run characterisation variation (about 4–15 ps/code).

(2) Dither fine-skew advisory

Not reliable on the board. Mechanism evidence chain:

1. Link dispersion: 32–48 sample pulse $\rightarrow$ 74–123 samples wide;
2. Window/period constraint: with a 130-sample period, the ± 64 window occupies 129 samples — no gap, no margin;

3. Window DC/slow-varying bias: A/B window means correlate with polarity; opposite edge biases of 100–240 ps;
4. Event-centre jumps: in-frame phase 44→9, spacing outliers;
5. Template/localisation sensitivity: same frame with different templates gives 0.72 ↔ 243 ps;
6. Tone-fit pollution: pulse duty cycle amplifies batch noise.

Fourteen methods were rejected with data (waveform geometry, window/gating sweeps, per-edge gain, detrending, two-parameter models, gap background estimation, median aggregation, event-centre refinement, gain-ratio calibration, etc.).

(3) Joint aggregation and differential measurement

- Joint cross-frame aggregation: board results 113–666 ps; occasional offline pass;
- Differential measurement (Δ dither vs Δ tone around a register step):
- Tone differential works in simulation (about +11.8 to +14.1 ps, expected +13.8 ps);
- Dither differential does not: dither valid frames drop from 1/10 to 0/10 on the simulated board.

3.6 Performance-Stage Errors

- A Stage-5 summary anomaly occurred: Mean RMSE 759, Mean correlation –0.998;
- Root cause: the canonical channel polarity was opposite to the actual signal;
- After the fix, board results recovered: Mean RMSE 23.15, Mean correlation +0.998354;
- The simulated board has no calibrated analog nonlinearity, jitter, or harmonic model, so SNDR/SFDR/ENOB are not used as numerical evidence.

4. Simulator Environment

4.1 What the simulator is

- A **simulated-board** environment that runs the **unmodified production firmware** on a host PC, with a virtual hardware model replacing the low-level board support package;
- The firmware's command parser, five-stage calibration flow, estimators, and DMA/SPI/UDP call paths all execute as real code;
- The virtual hardware model provides ADC/DAC behaviour, SPI registers, JESD state, DMA state machine, UDP bridge, and GPIO.

4.2 What the simulator models

- Virtual ADC signal chain: tone + dither sampling, channel gain/offset/skew, noise, pulse dispersion, sample-rate mismatch, saturation;
- Actuator model: fine-delay register writes change the inter-channel skew in real time;
- Ideal mode and simulated-board mode (dispersion + noise + skew drift);
- Fault injection: DMA errors, SPI timeouts, PLL unlock, etc.

4.3 What the simulator is used for

- Reproducing board behaviour without hardware: the simulated board reproduces "event detection valid, fine-skew unreliable";
- Rapid comparison of different dither structures (no-flat-top triangular, flat-top linear, PRBS, Golay);
- Validating estimator changes and failure modes without modifying production sources or using the board;
- Pre-screening and root-cause localisation before board experiments.

4.4 Limitations of the simulator

- It is not an RTL/transistor-level co-simulation; it does not cover the JESD electrical layer, detailed DMA timing, or cache coherency;
- It has no calibrated analog front-end model: no jitter, no ADC nonlinearity, no harmonics, no true SNDR/SFDR/ENOB;
- Pulse dispersion is a first-order proxy, not a physical electromagnetic model;
- Therefore simulator results are used for **relative comparison, failure-mode reproduction, and algorithm screening**, not as absolute hardware-accuracy predictions.

5. Simulation Evidence

5.1 Production Estimators on the Simulated Board (four dither structures)

Structure	Mode	dither_A valid	fine-skew valid
No-flat-top triangular	ideal	1.000	1.000
No-flat-top triangular	simulated board	1.000	0.008

Structure	Mode	dither_A valid	fine-skew valid
Flat-top linear 48/32	ideal	1.000	1.000
Flat-top linear 48/32	simulated board	1.000	0.015
PRBS	ideal/simulated board	0–0.013	0
Golay	ideal/simulated board	0/1.000	0/0.080

5.2 Dedicated Correlation Estimator (simulation)

- At the current operating point: Golay offset error ~0.3 codes, gain error ~2%;
- Fine-skew with multi-frame averaging: Golay ~**0.35 ps** (20 frames), PRBS ~2.1 ps;
- At high frequency (2.7 GSPS / 1200 MHz): tone remains accurate, but the code-based estimator degrades (Golay skew ~4 ps, gain ~3.6%).

5.3 Differential Measurement (simulated board)

- Tone differential: ~+11.8 to +14.1 ps (expected +13.8 ps);
- Dither differential: 0/10 valid frames on the simulated board; Δ not reliable.

6. Root-Cause Summary

Symptom	Root cause	Key evidence	Status / recommendation
Dither fine-skew unreliable	Link dispersion + window/period constraint + event jumps + template sensitivity	edge 100–1000 ps; fine-skew valid rate ~1% on simulated board	Current implementation unusable; coded sequences + dedicated estimator are the direction
Offset weak pass	Digital-template flat-top assumption invalid	"flat-top" samples are template-defined; 3–4 code deviation	Switch to polarity-balanced full-window averaging
Gain 0.24–0.44	Template-to-code mapping + dispersion + no constant ratio	flat ≈ full (0.249/0.235)	A/B relative estimation avoids the mapping problem
Stage-5 summary anomaly	Canonical polarity inverted	Mean RMSE 759 / corr –0.998	Fixed and verified on board

Symptom	Root cause	Key evidence	Status / recommendation
Tone primary <1 ps	LS phase difference + actuator loop	−0.87 ps, 2/2 pass	Current primary calibration mechanism
High- frequency accuracy uncertain	Jitter/bandwidth not modelled	2.7 GSPS/1200 MHz simulation still accurate	Real board requires front-end modelling

7. Conclusions and Recommendations

1. **The tone primary path is currently the most reliable:** <1 ps on board, Stage-5 parallel average 39.5 dB / 6.27 bits;
2. **Dither three-way separation is not usable in the current impulse + estimator + link combination:**
 - the offset flat-top assumption is invalid;
 - gain has no constant ratio;
 - skew is limited by dispersion and event localisation;
1. **If dither fine-skew is pursued:** PRBS/Golay with a dedicated estimator are the direction; in simulation Golay reaches ~0.35 ps with multi-frame averaging, clearly better than PRBS at ~2.1 ps;
2. **Short term:** keep the tone path as the calibration mechanism, and use dither only for event detection / structural verification;
3. **Medium term:** refine the Golay correlation/edge-aware estimator in simulation before porting to firmware;
4. **Engineering:** focus on the firmware pipeline's stage invalidation and estimator reliability, so that algorithmic improvements can be reproduced reliably in the production flow.